

Harsh Dave

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Education

Texas A&M University

M.S. Computer Science

College Station, TX

Expected 2028

Texas A&M University

B.S. Computer Science & B.S. Statistics, Summa Cum Laude

College Station, TX

May 2026 | GPA: 3.92/4.00

Research Interests

Computational genomics of *Mycobacterium tuberculosis*; molecular evolution and positive selection; tuberculosis–diabetes comorbidity; drug resistance.

Research Experience

Graduate Assistant Research (Computational Genomics)

June 2026 – Present

Ioerger Lab, Texas A&M University (Advisor: Thomas R. Ioerger)

College Station, TX

- Investigating evidence of positive selection in *M. tuberculosis* genomes from diabetes and non-diabetes patient cohorts, analyzing approximately **4,000 genes** to identify candidate genomic differences.
- Built reproducible **Python** and **Slurm** pipelines for cohort QC, FASTA processing, Bayesian dN/dS estimation with **GenomeMap**, and posterior comparisons across large-scale TAMU HPRC runs.
- Validated selection signals with pN/pS enrichment, chi-square testing, Benjamini-Hochberg and local-FDR correction, drug-resistance positive controls, and independent **PAML/codeml** site-model analyses.

Undergraduate Researcher

January 2025 – May 2025

UrbanResilience.AI Lab, Texas A&M University

College Station, TX

- Built **Python** ingestion and preprocessing pipelines integrating air-quality sensor APIs and web-scraped wildfire data across Los Angeles sources for downstream emergency-response analysis.

Manuscripts in Preparation

Dave, H., Shatby, A., and Ioerger, T. Differential Bayesian analysis of genomic sequences identifies *M. tuberculosis* genes under positive selection in TB patients with diabetes.

Teaching Experience

Teaching Assistant, CS 111

Spring 2026

Texas A&M University

College Station, TX

- Supported students in **Java** and object-oriented programming through labs, debugging help, and weekly assignment review.

Additional Experience

Software Engineering Capstone – Amazon-Sponsored

August 2025 – December 2025

Texas A&M University

College Station, TX

- Built a breach-intelligence analytics platform in a **3-person team**, implementing **Scrapy** and **Tor** ingestion plus a **LangChain** and **Gemini 2.5 Flash Lite** pipeline for entity extraction and threat classification.
- Designed storage and retrieval across **AWS S3**, **MongoDB**, **Elasticsearch**, **Redis**, and **Neptune**; delivered Streamlit and Kibana views for querying unstructured breach data by sector and geography.

AI Engineering Intern

May 2024 – August 2024

Videomagic

Remote

- Developed and evaluated **PyTorch** and **Hugging Face Transformers** computer-vision workflows for AI-generated video detection, analyzing frame-level outputs and model failure modes.
- Built **Feathers.js**, **MySQL**, and **Knex.js** APIs for model inference and data management, secured with **Auth0** and **JWT RBAC**.

Technical Skills

Data Science & Statistics: Python, R, SQL, pandas, NumPy, SciPy, scikit-learn, Bayesian inference, MCMC, hypothesis testing, FDR correction, Matplotlib

Bioinformatics & HPC: GenomeMap, PAML/codeml, dN/dS, pN/pS, FASTA pipelines, Slurm, Linux, TAMU HPRC

Machine Learning & Data Systems: PyTorch, Hugging Face Transformers, LangChain, AWS, MongoDB, Elasticsearch, Redis, Git, GitHub Actions

Programming: Java, C++, TypeScript, JavaScript, React, FastAPI, Docker, PostgreSQL